www.tntextbooks.in

4. Measurement of growth

Activity

Measurement of growth by direct method.

Step 1: Take ordinary scale.

Step 2: Measure ground stem up to the growing point of the plant.

Step 3: Use Indian ink and mark at regular intervals to measure the length of root, stem, and girth of the trunk.

5. Sequence of developmental process in a plant cell

Development is a term that includes all the changes that an organism goes through during life cyle from germination of a seed to senescence. Diagrammatic representation of the sequence of processes which constitute the development of a cell of a higher plant is given in the figure. It is also applicable to tissues/organ.

Experiment: 1. Arc auxanometer:

The increase in the length of the stem tip can easily be measured by an arc auxanometer which consists of a small pulley to the axis of which is attached a long pointer sliding over a graduated arc. A thread one end of which is tied to the stem tip and another end to a weight passes over the pulley tightly. As soon as the stem tip increases in length, the pulley moves and the pointer slide over the graduated arc (Figure 15.8). The reading is taken. The actual increase in the length of the stem is then calculated by knowing the length of the pointer and the radius of the pulley. If the radius of the pulley is 4 inches and the length of pointer 20 inches the actual growth is measured as follows:

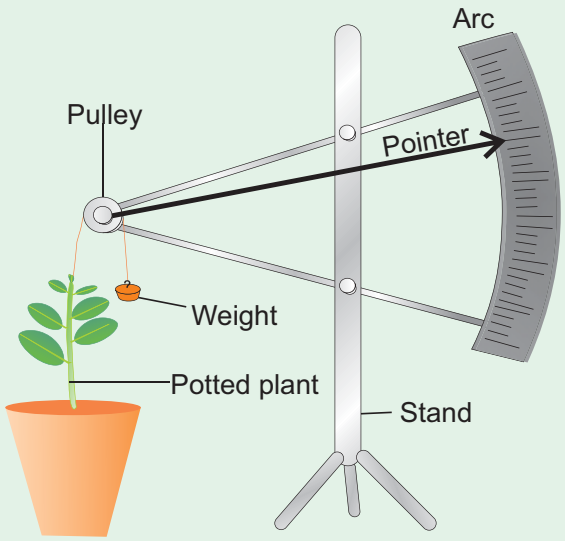


Figure 15.8: Arc auxanometer

$$\text{Actual growth in length} = \frac{\text{Distance travelled by the pointer} \times \text{radius of the pulley}}{\text{Length of the pointer}}$$

$$\text{For example, actual growth in length} = \frac{10 \times 4 \text{ inches}}{20 \text{ inches}}$$

$$= 2 \text{ inches}$$